CHE 201 Spring 2020 Quiz 2

Roxanna Mendoza

TOTAL POINTS

97 / 100

QUESTION 1

1 40 pts

1.1 1a **20** / **20**

✓ **- 0 pts** **Correct**

- 5 pts not explained

1.2 1b **20** / **20**

✓ **- 0 pts** **Correct**

- 5 pts v1 and v2 are incorrect

- 5 pts state 1 not shown

QUESTION 2

2 30 pts

2.1 2a **15** / **15**

✓ **- 0 pts** **Correct**

- 2 pts calculation error

2.2 2b **15** / **15**

✓ **- 0 pts** **Correct**

- 2 pts calculation error

QUESTION 3

3 3 **27** / **30**

- 0 pts Correct

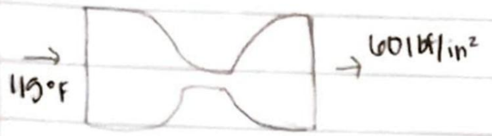
- 3 pts not explained

✓ **- 3 pts** **calculation error, Q is -2kj/s**

Roxanna Mendoza UIN: 652739964

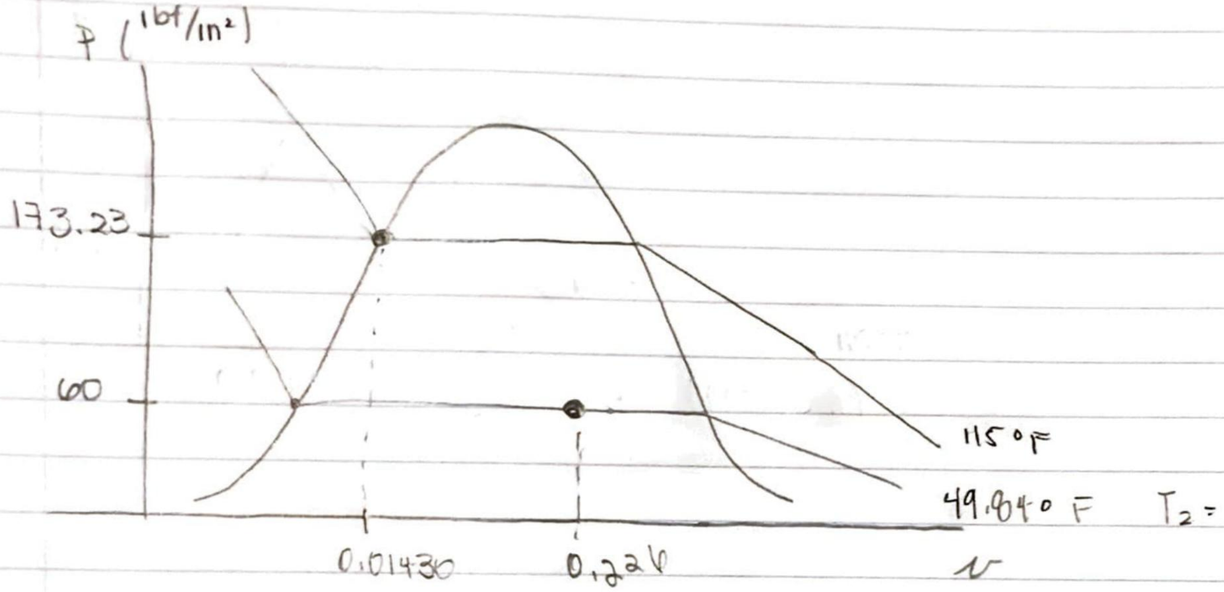
Quiz #2

#1. Sketch:



Engineering Model:

1. open system
2. $\Delta KE = \Delta PE = 0$
3. $Q = 0$



$h_{f_1} = 50.512$ Btu/lbm

@ $P = 60$ lbf/in²

$h_{f_2} = 28.080$ $h_{g_2} = 110.11$ $v_f = 0.01270$ $v_g = 0.79361$

$h_1 = h_{f2} + x_2(h_{g2} - h_{f2})$

$x_2 = \dfrac{(50.512) - (28.080)}{(110.11) - (28.080)} = 0.273 \times 100 = 27.3\%$

$T_2 = 49.84°F$

$v_2 = 0.01270 + (0.273)(0.79361 - 0.01270) =$

$v_2 = 0.226$ ft³/lbm

1.1 1a **20 / 20**

✓ **- 0 pts** Correct

- 5 pts not explained

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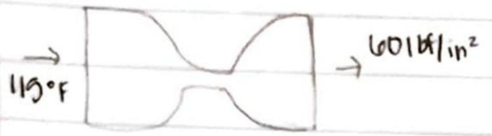

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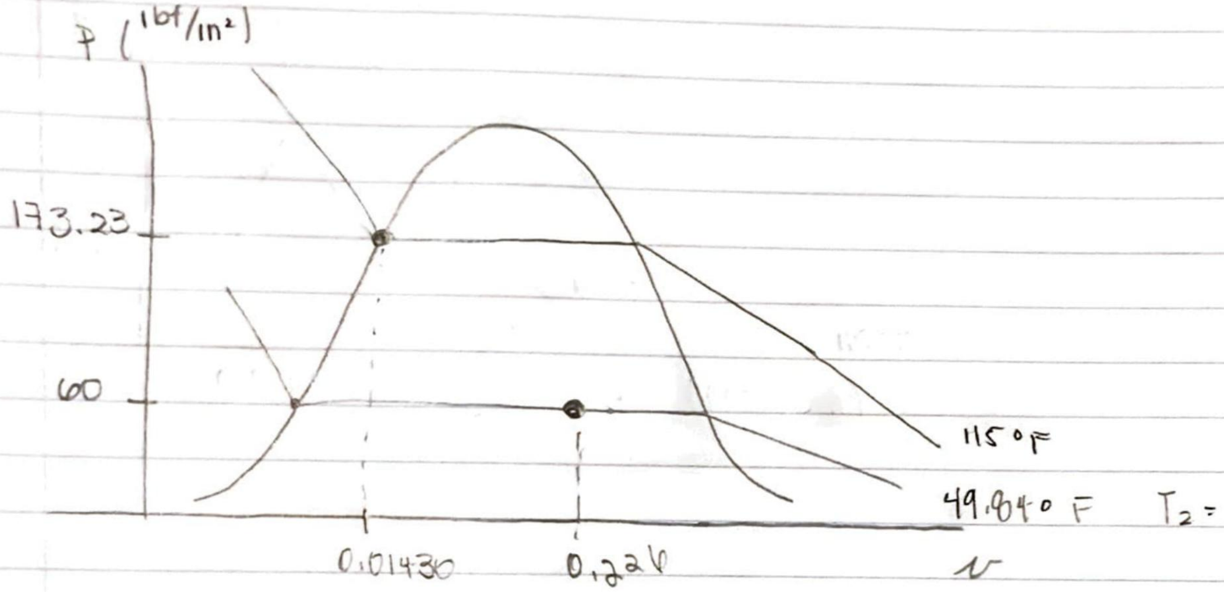


$h_{f_1} = 50.512$ Btu/lbm

@ $P = 60$ lbf/in²

$h_{f_2} = 28.080$ $h_{g_2} = 110.11$ $v_f = 0.01270$ $v_g = 0.79361$

$$h_1 = h_{f2} + x_2 (h_{g2} - h_{f2})$$

$$x_2 = \frac{(50.512) - (28.080)}{(110.11) - (28.080)} = 0.273 \times 100 = 27.3\%$$

$T_2 = 49.840$ °F

$v_2 = 0.01270 + (0.273)(0.79361 - 0.01270) =$

$v_2 = 0.226$ ft³/lbm

1.2 1b **20 / 20**

✓ **- 0 pts** Correct

- 5 pts v1 and v2 are incorrect

- 5 pts state 1 not shown

#2 Sketch:

Engineering Model:
① Open system
② ideal gas

Given:

$\bar{V}\cdot A = \frac{85\ L}{min} \Big| \frac{0.001\ m^3}{1\ L} \Big| \frac{1\ min}{60\ sec} = 0.00142\ \frac{m^3}{s}$

$T = 22°C + 273.15 = 295.15\ K$

$P = 3.6\ bar \Big| \frac{100000\ Pa}{1\ bar} = 360{,}000\ Pa$

$d = 9\ cm = 0.090\ m \qquad r = 0.04\ m$

$A = \pi r^2 = \pi(0.04)^2 = 0.005\ m^2$

$\frac{0.00142}{0.005} = \bar{V} \qquad \boxed{\bar{V} = 0.282\ m/s}$ ← velocity

Part B:

$\dot{m} = \frac{\bar{V}\cdot A}{v} \qquad v = \frac{RT_1}{P_1} \qquad v_{specific} = \frac{v_1}{M}$

$v_{specific} = \frac{\frac{(8314\ \frac{N\cdot m}{K})(295.15\ K)}{(360{,}000\ Pa)}}{(28.97\ kg)} = 0.235\ \frac{m^3}{kg}$

$\dot{m} = \frac{(0.00142\ m^3/sec)}{(0.235\ m^3/kg)} = \boxed{0.006\ \frac{kg}{sec}}$ ← mass flow rate

2.1 2a **15 / 15**

✓ **- 0 pts** Correct

- 2 pts calculation error

#2 Sketch:

Engineering Model:
1. Open system
2. ideal gas

Given:

$$\dot{V} \cdot A = \frac{85\ L}{min} \left| \frac{0.001\ m^3}{1\ L} \right| \frac{1\ min}{60\ sec} = 0.00142\ \frac{m^3}{s}$$

$$T = 22°C + 273.15 = 295.15\ K$$

$$P = 3.6\ bar \left| \frac{100000\ Pa}{1\ bar} \right. = 360{,}000\ Pa$$

$d = 8\ cm = 0.080\ m$ $\quad$ $r = 0.04\ m$

$$A = \pi r^2 = \pi (0.04)^2 = 0.005\ m^2$$

velocity

$$\frac{0.00142}{0.005} = V \qquad V = 0.282\ m/s$$

Part B:

$$\dot{m} = \frac{V \cdot A}{v} \qquad v = \frac{RT_1}{P_1} \qquad v_{specific} = \frac{v_1}{M}$$

$$v_{specific} = \frac{\frac{(8314\ \frac{N \cdot m}{K})(295.15\ K)}{(360{,}000\ Pa)}}{(28.97\ kg)} = 0.235\ \frac{m^3}{kg}$$

$$\dot{m} = \frac{(0.00142\ m^3/sec)}{(0.235\ m^3/kg)} = 0.006\ \frac{kg}{sec}$$

mass flow rate

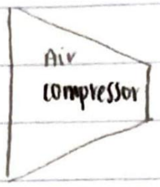
2.2 2b **15 / 15**

✓ **- 0 pts** Correct

- 2 pts calculation error

#3.

Sketch:



Engineering model:

① open system
② ideal gas
③ $\Delta PE = \Delta KE = 0$
④ steady state

Given:

$$P_1 = \frac{1.05\text{ bar}}{} \left| \frac{1\times10^5\text{ Pa}}{1\text{ bar}} \right. = 105000\text{ Pa}$$

$T_1 = 300\text{K}$

$A \cdot V = 12\,m^3/min \div 60\text{ sec} = 0.2\ m^3/sec$

$$P_2 = \frac{12\text{ bar}}{} \left| \frac{1\times10^5\text{ Pa}}{1\text{ bar}} \right. = 1{,}200{,}000\text{ Pa}$$

$T_2 = 400\text{ K}$

$Q = 2\text{ kW} = 2\text{ kJ/sec}$

$\dot{W}_{cv} = \dot{Q}_{cv} + \dot{m}(h_1 - h_2)$

$$v_{specific} = \frac{(8.314\ \frac{N\cdot m}{K})(300\text{ K})}{\frac{(105000\text{ Pa})}{28.97\text{ kg}}} = 0.82\ \frac{m^3}{kg}$$

$$\dot{m} = \frac{0.2\ m^3/sec}{0.82\ m^3/kg} = 0.244\ \frac{kg}{sec}$$

@ $T_1 = 300\text{ K}$ $h_1 = 300.19$
@ $T_2 = 400\text{K}$ $h_2 = 400.98$

$$\dot{W}_{cv} = 2\ \frac{kJ}{sec} + \left(0.244\ \frac{kg}{sec}\right)(300.19 - 400.98) = -22.58\text{ kW}$$

3 3 **27** / **30**

- **- 0 pts** Correct
- **- 3 pts** not explained
- ✓ **- 3 pts** calculation error, Q is -2kj/s